RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 4 (Lambda)
Topic(s): Calculus II
(Integration, Applications, DE) (Solutions)

Source of Question: TJC Prelim 9758/2022/01/Q1

1 Find

(a) $\int \sin^{-1} 2x \, \mathrm{d}x$, [3]

(b) $\int \frac{1}{x^2 - nx + n^2} \, \mathrm{d}x$, where n is a nonzero constant. [2]

Solution:

(i)

$$\int \sin^{-1} 2x \, \mathrm{d}x = \int 1.\sin^{-1} 2x \, \mathrm{d}x$$

$u = \sin^{-1} 2x$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2$ $\quad v = x$

$$= x\sin^{-1} 2x - \int x.\frac{2}{\sqrt{1-4x^2}} \, \mathrm{d}x$$

$$= x\sin^{-1} 2x + \frac{1}{4}\int (-8x)\left(1-4x^2\right)^{-\frac{1}{2}} \, \mathrm{d}x$$

$$= x\sin^{-1} 2x + \frac{1}{4}.\frac{\left(1-4x^2\right)^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= x\sin^{-1} 2x + \frac{1}{2}\sqrt{1-4x^2} + c$$

(ii)

$$\int \frac{1}{x^2 - nx + n^2} \, \mathrm{d}x = \int \frac{1}{\left(x - \frac{1}{2}n\right)^2 - \frac{1}{4}n^2 + n^2} \, \mathrm{d}x$$

$$= \int \frac{1}{\left(x - \frac{1}{2}n\right)^2 + \frac{3}{4}n^2} \, \mathrm{d}x$$

$$= \frac{2}{\sqrt{3}n} \tan^{-1}\left(\frac{x - \frac{1}{2}n}{\frac{\sqrt{3}}{2}n}\right) + c$$

$$= \frac{2}{\sqrt{3}n} \tan^{-1}\left(\frac{2x-n}{\sqrt{3}n}\right) + c$$

Source of Question: TMJC Prelim 9758/2022/01/Q2(modified)

2 **(a)** Find $\displaystyle\int \frac{1}{\sqrt{\left(1-x^2\right)\sin^{-1}x}}\,\mathrm{d}x$. [2]

(b) Find $\displaystyle\int \frac{x-3}{x^2+2x-3}\,\mathrm{d}x$. [3]

Solution:

(a)

$$\int \frac{1}{\sqrt{\left(1-x^2\right)\sin^{-1}x}}\,\mathrm{d}x = \int \frac{1}{\sqrt{\left(1-x^2\right)}}\left(\sin^{-1}x\right)^{-\frac{1}{2}}\,\mathrm{d}x$$

$$= \frac{\left(\sin^{-1}x\right)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 2\sqrt{\sin^{-1}x} + C$$

The integrand is of the form $\mathrm{f}'(x)(\mathrm{f}(x))^n$, where $\mathrm{f}(x) = \sin^{-1}x$ and $n = -\frac{1}{2}$.

(b)

$$\int \frac{x-3}{x^2+2x-3}\,\mathrm{d}x = \int \frac{x-3}{(x-1)(x+3)}\,\mathrm{d}x$$

$$= -\frac{1}{2}\int \frac{1}{x-1}\,\mathrm{d}x + \frac{3}{2}\int \frac{1}{x+3}\,\mathrm{d}x$$

$$= -\frac{1}{2}\ln|x-1| + \frac{3}{2}\ln|x+3| + C$$

Alternatively,

$$\int \frac{x-3}{x^2+2x-3}\,\mathrm{d}x = \frac{1}{2}\int \frac{2x+2}{x^2+2x-3}\,\mathrm{d}x - \int \frac{4}{x^2+2x-3}\,\mathrm{d}x$$

$$= \frac{1}{2}\ln\left|x^2+2x-3\right| - 4\int \frac{1}{(x+1)^2-4}\,\mathrm{d}x$$

$$= \frac{1}{2}\ln|(x+3)(x-1)| - \frac{4}{4}\ln\left|\frac{x+1-2}{x+1+2}\right| + C$$

$$= \frac{1}{2}\ln|x+3| + \frac{1}{2}\ln|x-1| - \left(\ln|x-1| - \ln|x+3|\right) + C$$

$$= -\frac{1}{2}\ln|x-1| + \frac{3}{2}\ln|x+3| + C$$

Source of Question: NYJC Prelim 9758/2022/02/Q5 (modified)

3 **(a)** **(i)** Differentiate $\sqrt{x^2+4}$ with respect to x. [1]

(ii) Hence find $\displaystyle\int \frac{x^3}{\sqrt{x^2+4}}\,\mathrm{d}x$. [3]

(b) Find **(i)** $\displaystyle\int \sin^2 mx\,\mathrm{d}x$, where m is a non-zero constant, [2]

(ii) $\displaystyle\int \cos^3 3x\,\mathrm{d}x$. [3]

Solution:

(a)(i)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{x^2+4}\right)=\frac{2x}{2\sqrt{x^2+4}}=\frac{x}{\sqrt{x^2+4}}$$

(a)(ii)

$$u=x^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x}=\frac{x}{\sqrt{x^2+4}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x}=2x \qquad v=\sqrt{x^2+4}$$

$$\int \frac{x^3}{\sqrt{x^2+4}}\,\mathrm{d}x = x^2\sqrt{x^2+4}-\int 2x\sqrt{x^2+4}\,\mathrm{d}x$$

$$= x^2\sqrt{x^2+4}-\frac{2\left(x^2+4\right)^{\frac{3}{2}}}{3}+C$$

(b)(i)

$$\int \sin^2 mx\,\mathrm{d}x=\frac{1}{2}\int 1-\cos 2mx\,\mathrm{d}x$$

$$=\frac{1}{2}x-\frac{\sin 2mx}{4m}+C$$

(b)(ii)

$$\int \cos^3 3x\,\mathrm{d}x=\int \cos 3x\left(\cos^2 3x\right)\mathrm{d}x$$

$$=\int \cos 3x\left(1-\sin^2 3x\right)\mathrm{d}x$$

$$=\int \cos 3x-\cos 3x\sin^2 3x\,\mathrm{d}x$$

$$=\frac{\sin 3x}{3}-\frac{\sin^3 3x}{9}+C$$

Source of Question: NJC Prelim 9758/2022/01/Q1

4 **(i)** Given that $I_n = \int_1^{\mathrm{e}} x(\ln x)^n \,\mathrm{d}x$ for $n \in \mathbb{Z},\ n \geq 0,$ show that

$$I_n = \frac{\mathrm{e}^2}{2} - \frac{n}{2} I_{n-1}$$

for all $n \in \mathbb{Z}^+$. [2]

(ii) Find the exact volume of the solid generated when the region bounded by the curve $y = \sqrt{x}\ln x,$ the x-axis and the line $x = \mathrm{e}$ is rotated completely about the x-axis. [3]

Solution:

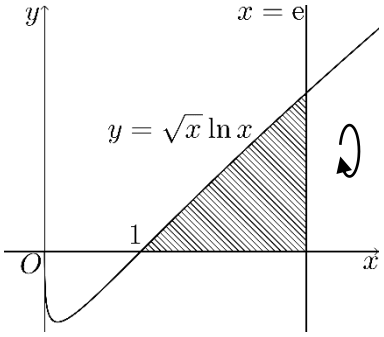
(i)

$$\begin{aligned} I_n &= \int_1^{\mathrm{e}} x(\ln x)^n \,\mathrm{d}x \\ &= \left[\frac{x^2}{2}(\ln x)^n\right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} \frac{x^2}{2} \cdot n(\ln x)^{n-1} \frac{1}{x} \,\mathrm{d}x \\ &= \frac{\mathrm{e}^2}{2} - \frac{n}{2}\int_1^{\mathrm{e}} x(\ln x)^{n-1} \,\mathrm{d}x \\ &= \frac{\mathrm{e}^2}{2} - \frac{n}{2} I_{n-1} \text{ (shown)} \end{aligned}$$

(ii)

Volume

$$\begin{aligned} &= \pi\int_1^{\mathrm{e}} x(\ln x)^2 \,\mathrm{d}x \\ &= \pi I_2 \\ &= \pi\left(\frac{\mathrm{e}^2}{2} - \frac{2}{2} I_1\right) \\ &= \pi\left[\frac{\mathrm{e}^2}{2} - \left(\frac{\mathrm{e}^2}{2} - \frac{1}{2} I_0\right)\right] \\ &= \frac{\pi}{2} I_0 = \frac{\pi}{2}\left[\frac{x^2}{2}\right]_1^{\mathrm{e}} \\ &= \frac{\pi}{4}\left(\mathrm{e}^2 - 1\right) \text{ units}^3 \end{aligned}$$



Alternatively:

Volume

$$= \pi \int_1^{\mathrm{e}} x(\ln x)^2 \, \mathrm{d}x$$

$$= \pi \left\{ \left[\frac{x^2}{2}(\ln x)^2 \right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} \frac{x^2}{2} \cdot 2(\ln x) \frac{1}{x} \mathrm{d}x \right\}$$

$$= \pi \left[\frac{\mathrm{e}^2}{2} - \int_1^{\mathrm{e}} x(\ln x) \mathrm{d}x \right]$$

$$= \pi \left\{ \frac{\mathrm{e}^2}{2} - \left[\frac{x^2}{2}(\ln x) \right]_1^{\mathrm{e}} + \int_1^{\mathrm{e}} \frac{x^2}{2} \cdot \frac{1}{x} \mathrm{d}x \right\}$$

$$= \pi \left\{ \frac{\mathrm{e}^2}{2} - \frac{\mathrm{e}^2}{2} + \left[\frac{x^2}{4} \right]_1^{\mathrm{e}} \right\}$$

$$= \frac{\pi}{4}(\mathrm{e}^2 - 1) \text{ units}^3$$

$u = (\ln x)^2$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2(\ln x)\frac{1}{x}$ $\quad v = \frac{x^2}{2}$

$u = \ln x$ $\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ $\quad v = \frac{x^2}{2}$

Source of Question: ACJC Prelim 9758/2022/01/Q4

5 (a) The continuous function $\mathrm{f}(x)$, where $\mathrm{f}(x) > 0$, is strictly decreasing for $x \geq 1$.
Sketch the curve $y = \mathrm{f}(x)$ for $k < x < k+1$ where k is an integer and $k \geq 1$.
By comparing the areas of appropriate rectangles and the area under the curve $y = \mathrm{f}(x)$, show that for any integer $k \geq 1$,

$$\mathrm{f}(k+1) < \int_k^{k+1} \mathrm{f}(x)\,\mathrm{d}x < \mathrm{f}(k).$$ [2]

(b) The region under the curve $y = \dfrac{1}{x}$ between $x = 1$ and $x = 10$, is split into 9 vertical strips of equal width. Use the result in part **(a)** to prove

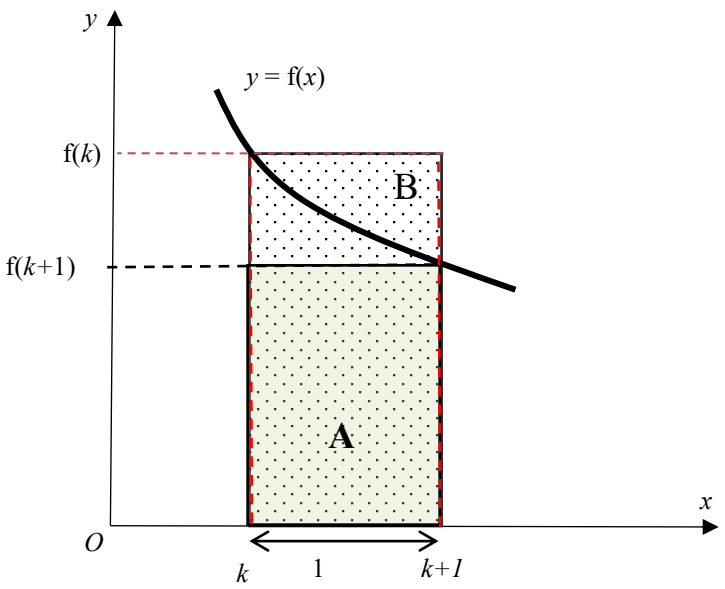
(i) $\displaystyle\int_1^{10} \frac{1}{x}\,\mathrm{d}x < \sum_{k=1}^{9} \frac{1}{k}$, [1]

(ii) $\displaystyle\sum_{k=1}^{9} \frac{1}{k} < 1 + \int_1^{9} \frac{1}{x}\,\mathrm{d}x$. [2]

Hence show that $\ln 10 < 1 + \dfrac{1}{2} + \dfrac{1}{3} + ... + \dfrac{1}{9} < 1 + \ln 9$. [1]

Solution:

(a)



Area under curve = $\displaystyle\int_k^{k+1} \mathrm{f}(x)\,\mathrm{d}x$

Area of rectangle A= $\mathrm{f}(k+1)$ x 1

Area of rectangle B= $\mathrm{f}(k)$ x 1

As seen from diagram:

$$\mathrm{f}(k+1) < \int_k^{k+1} \mathrm{f}(x)\,\mathrm{d}x < \mathrm{f}(k)$$

(b)(i)

Let $f(x)=\frac{1}{x}$. Then,

$$\int_1^{10}\frac{1}{x}\,dx=\int_1^{2}\frac{1}{x}\,dx+\int_2^{3}\frac{1}{x}\,dx+\int_3^{4}\frac{1}{x}\,dx+....+\int_9^{10}\frac{1}{x}\,dx$$

$$<f(1)+f(2)+f(3)+.....+f(9)$$

$$=\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{9}$$

$$=\sum_{k=1}^{9}\frac{1}{k}$$

(b)(ii)

$$\int_1^{9}\frac{1}{x}\,dx=\int_1^{2}\frac{1}{x}\,dx+\int_2^{3}\frac{1}{x}\,dx+\int_3^{4}\frac{1}{x}\,dx+....+\int_8^{9}\frac{1}{x}\,dx$$

$$>f(2)+f(3)+f(4)+.....+f(9)$$

$$=\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{9}=\sum_{k=1}^{9}\frac{1}{k}-1$$

$$1+\int_1^{9}\frac{1}{x}\,dx>\sum_{k=1}^{9}\frac{1}{k}$$

$$\therefore \sum_{k=1}^{9}\frac{1}{k}<1+\int_1^{9}\frac{1}{x}\,dx$$

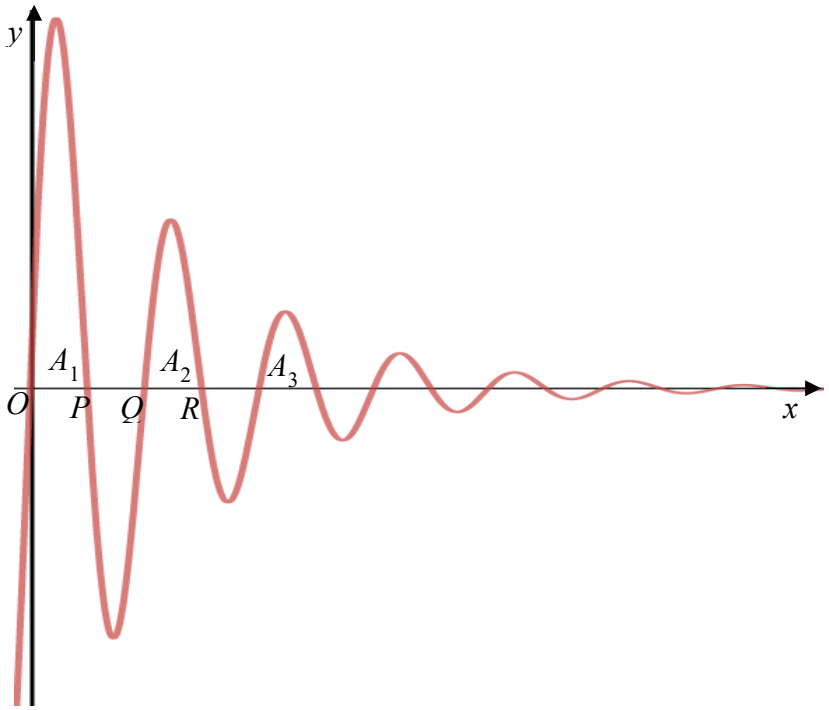
$$\int_1^{10}\frac{1}{x}\,dx<\sum_{k=1}^{9}\frac{1}{k}<1+\int_1^{9}\frac{1}{x}\,dx$$

$$\left[\ln x\right]_1^{10}<1+\frac{1}{2}+\frac{1}{3}+.....\frac{1}{9}<1+\left[\ln x\right]_1^{9}$$

$$\ln 10<1+\frac{1}{2}+\frac{1}{3}+.....\frac{1}{9}<1+\ln 9$$

Source of Question: EJC JC2 Mid-Year CT 9758/2018/01/Q12

6



The figure shows part of the curve C. It illustrates a damped oscillation in which the amplitude of oscillation decreases with time, for instance when frictional forces diminish the amplitude of oscillation of a spring until it comes to rest. It is given that curve C has equation

$$y = \mathrm{e}^{-x}\sin 2x, x \geq 0.$$

(i) Show that the coordinates where C cuts the positive x-axis are $(\frac{k}{2}\pi, 0)$, where $k \in \mathbb{Z}^+$. [2]

(ii) Use integration by parts to show that

$$\int \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x = -\frac{\mathrm{e}^{-x}}{5}\left(\sin 2x + 2\cos 2x\right) + c \text{, where } c \text{ is an arbitrary constant} \quad [4]$$

(iii) The terms of the sequence $A_1, A_2, A_3, \ldots, A_n, \ldots$ represent areas between C and the x-axis for successive portions of C where y is positive. The area represented by A_1, A_2 and A_3 are shown in the figure. It is given that $A_n = \int_{(n-1)\pi}^{(2n-1)\pi/2} \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x$. Using the results in (ii), show that $A_n = \frac{2}{5}\mathrm{e}^{-n\pi}\left(\mathrm{e}^{\pi} + \mathrm{e}^{\frac{\pi}{2}}\right)$. [4]

(iv) Show that $A_1, A_2, A_3, \ldots, A_n, \ldots$ is a geometric progression.

Hence prove that $\sum_{n=1}^{\infty} A_n = \frac{2}{5(1-\mathrm{e}^{-\frac{\pi}{2}})}$. [4]

Solution:

(i)

$e^{-x}\sin 2x = 0 \Rightarrow \sin 2x = 0 \; [\, e^{-x} > 0 \,]$

$\Rightarrow 2x = k\pi \qquad k \in \mathbb{Z}^+$

$\Rightarrow x = \dfrac{k\pi}{2} \qquad k \in \mathbb{Z}^+$

(ii)

$$\int \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x = -\mathrm{e}^{-x}\sin 2x + 2\int \mathrm{e}^{-x}\cos 2x \,\mathrm{d}x$$

$$= -\mathrm{e}^{-x}\sin 2x + 2\left\{-\mathrm{e}^{-x}\cos 2x - 2\int \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x\right\}$$

$$= -\mathrm{e}^{-x}\sin 2x - 2\mathrm{e}^{-x}\cos 2x - 4\int \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x$$

$$\Rightarrow 5\int \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x = -\mathrm{e}^{-x}\left(\sin 2x + 2\cos 2x\right)$$

$$\Rightarrow \int \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x = -\frac{\mathrm{e}^{-x}}{5}\left(\sin 2x + 2\cos 2x\right) + c$$

where c is an arbitrary constant.

(iii)

$$A_n = \int_{(n-1)\pi}^{(2n-1)\pi/2} \mathrm{e}^{-x}\sin 2x \,\mathrm{d}x = \left[-\frac{\mathrm{e}^{-x}}{5}\left(\sin 2x + 2\cos 2x\right)\right]_{(n-1)\pi}^{(2n-1)\pi/2}$$

$$= -\frac{\mathrm{e}^{-(2n-1)\pi/2}}{5}\left(\sin\left(2n-1\right)\pi + 2\cos\left(2n-1\right)\pi\right)$$

$$+ \frac{\mathrm{e}^{-(n-1)\pi}}{5}\left(\sin 2\left(n-1\right)\pi + 2\cos 2\left(n-1\right)\pi\right)$$

$$= -\frac{\mathrm{e}^{-n\pi+\pi/2}}{5}\left(0 + 2\left(-1\right)\right) + \frac{\mathrm{e}^{-n\pi+\pi}}{5}\left(0 + 2\left(1\right)\right)$$

$$= \frac{\mathrm{e}^{-n\pi}}{5}\left(2\mathrm{e}^{\pi/2} + 2\mathrm{e}^{\pi}\right) = \frac{2}{5}e^{-n\pi}\cdot\left(e^{\pi} + e^{\frac{\pi}{2}}\right)$$

(iv)

$$A_n = \frac{2}{5}e^{-n\pi}\cdot\left(e^{\pi} + e^{\frac{\pi}{2}}\right) \Rightarrow A_{n-1} = \frac{2}{5}e^{-(n-1)\pi}\cdot\left(e^{\pi} + e^{\frac{\pi}{2}}\right)$$

Since $\dfrac{A_n}{A_{n-1}} = e^{-\pi}$ is a constant, $A_1, A_2, A_3, \ldots, A_n, \ldots$ is a geometric progression.

$$\sum_{n=1}^{\infty} A_n = \frac{\frac{2}{5}\mathrm{e}^{-\pi}\left(e^{\pi} + e^{\frac{\pi}{2}}\right)}{\left(1-\mathrm{e}^{-\pi}\right)} = \frac{\frac{2}{5}\left(1 + e^{-\frac{\pi}{2}}\right)}{\left(1-\mathrm{e}^{-\pi}\right)} = \frac{2\left(1 + e^{-\frac{\pi}{2}}\right)}{5\left(1 + e^{-\frac{\pi}{2}}\right)\left(1 - e^{-\frac{\pi}{2}}\right)} = \frac{2}{5(1-\mathrm{e}^{-\frac{\pi}{2}})}$$ (Shown)

Source of Question: CJC Prelim 9758/2022/02/Q5

7 A curve C_1 has equation $y = \mathrm{f}(x)$, where $k > 1$ and

$$\mathrm{f}(x) = \begin{cases} x + k & \text{for } -k < x \le 0, \\ (k+1) - \dfrac{(2k-x)^2}{4k^2} & \text{for } 0 < x \le 2k, \\ 0 & \text{otherwise.} \end{cases}$$

(i) Sketch the curve C_1. [2]

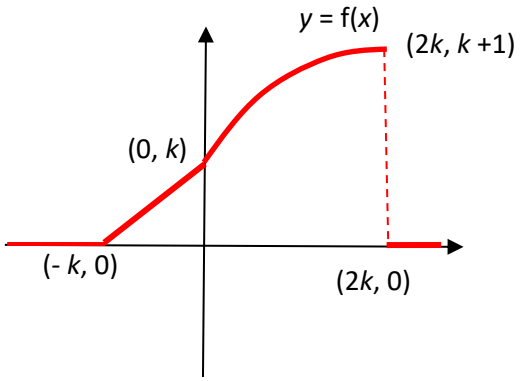
(ii) The region R is bounded by the curve C_1, the line $x = 1$ and the x-axis. Find the exact area of R, leaving your answer in terms of k. [6]

(iii) On the same diagram on part **(i)**, sketch the curve C_2 with equation $y = x^2 \tan^{-1} x$. [1]

(iv) The region S is bounded by the curves C_1 and C_2, and the x-axis. Given that $k = 3$, determine the volume of solid of revolution formed when S is rotated 2π radians about the x-axis, leaving your answer to 2 decimal places. [3]

Solution:

(i)



(ii)

Required Area

$$= \int_{-k}^{0} (x+k)\,dx + \int_{0}^{1} (k+1) - \frac{(2k-x)^2}{4k^2}\,dx$$

$$= \left[\frac{x^2}{2} + kx\right]_{-k}^{0} + \left[(k+1)x - \frac{(2k-x)^3}{4k^2(3)(-1)}\right]_{0}^{1}$$

$$= 0 - \left[\frac{k^2}{2} - k^2\right] + \left[(k+1) - \frac{(2k-1)^3}{4k^2(3)(-1)}\right] - \left[-\frac{(2k)^3}{4k^2(3)(-1)}\right]$$

$$= \frac{k^2}{2} + \frac{k}{3} + 1 + \frac{(2k-1)^3}{12k^2}$$

OR

Required Area

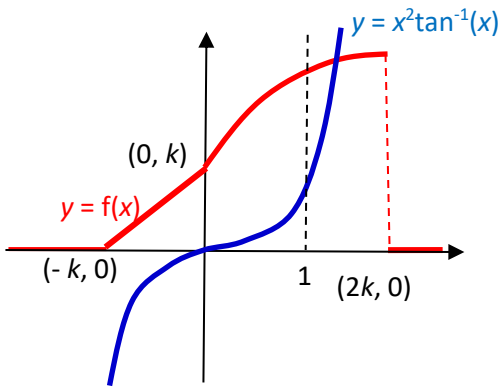
$$= \frac{1}{2}(k)(k) + \int_{0}^{1} (k+1) - \frac{(2k-x)^2}{4k^2}\,dx$$

$$= \frac{k^2}{2} + \left[(k+1)x - \frac{(2k-x)^3}{4k^2(3)(-1)} \right]_0^1$$

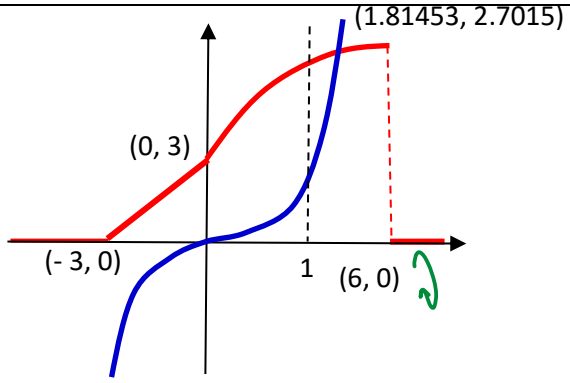
$$= \frac{k^2}{2} + \left[(k+1) - \frac{(2k-1)^3}{4k^2(3)(-1)} \right] - \left[-\frac{(2k)^3}{4k^2(3)(-1)} \right]$$

$$= \frac{k^2}{2} + \frac{k}{3} + 1 + \frac{(2k-1)^3}{12k^2}$$

(iii)



(iv)



Point of intersection between C_1 and C_2 is at $x = 1.81453$

Volume of S

$$= \pi \left\{ \int_{-3}^{0} (x+3)^2 dx + \int_{0}^{1.81453} \left[(3+1) - \frac{(2(3)-x)^2}{4(3)^2} \right]^2 dx - \int_{0}^{1.81453} \left[x^2 \tan^{-1}(x) \right]^2 dx \right\}$$

= 77.54 (2 d.p.) unit^3

OR

Volume of S

$$= \frac{1}{3}\pi(3)^2(3) + \pi \left\{ \int_{0}^{1.81453} \left[(3+1) - \frac{(2(3)-x)^2}{4(3)^2} \right]^2 dx - \int_{0}^{1.81453} \left[x^2 \tan^{-1}(x) \right]^2 dx \right\}$$

= 77.54 (2 d.p.) unit^3

Source of Question: DHS Prelim 9758/2025/01/Q1

8 It is given that $2xy\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 - y^2$ where $x > 0,\ y < 0$. Using the substitution $u = \dfrac{y}{x}$, show that the differential equation can be transformed to $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1-3u^2}{2ux}$. Hence find the general solution of y in terms of x. [6]

Solution:

$u = \dfrac{y}{x} \Rightarrow y = ux \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}u}{\mathrm{d}x}x + u$ ... (1)

$2xy\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 - y^2$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2 - y^2}{2xy} = \dfrac{x^2 - u^2x^2}{2ux^2} = \dfrac{1-u^2}{2u}$ ... (2)

Equate (1) with (2):

$\dfrac{\mathrm{d}u}{\mathrm{d}x}x + u = \dfrac{1-u^2}{2u}$

$\dfrac{\mathrm{d}u}{\mathrm{d}x}x = \dfrac{1-u^2}{2u} - u = \dfrac{1-3u^2}{2u}$

$\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1-3u^2}{2ux}$ (shown)

$\displaystyle\int \frac{2u}{1-3u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x$

$\displaystyle -\frac{1}{3}\int \frac{-6u}{1-3u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x$

$-\dfrac{1}{3}\ln\left|1-3u^2\right| = \ln x + c \quad (\because x > 0)$

$\ln\left|1-3u^2\right| = -3\ln x - 3c$

$1-3u^2 = \pm \mathrm{e}^{-3c}\cdot \mathrm{e}^{\ln\frac{1}{x^3}} = \dfrac{A}{x^3}$ (where $A = \pm\mathrm{e}^{-3c}$)

$3u^2 = 1 - \dfrac{A}{x^3}$

$\left(\dfrac{y}{x}\right)^2 = \dfrac{1}{3}\left(1 - \dfrac{A}{x^3}\right)$

$y = -\sqrt{\dfrac{x^2}{3}\left(1 - \dfrac{A}{x^3}\right)} = -x\sqrt{\dfrac{1}{3}\left(1 - \dfrac{A}{x^3}\right)} \quad (\because y < 0)$

Source of Question: TJC Prelim 9758/2021/01/Q10

9 On a mangrove swamp, scientists are investigating the population of mudskippers and crabs. Initially, there are 800 mudskippers and 800 crabs on the mangrove swamp. At time t years, the number of mudskippers and crabs are M and C respectively.

(a) For the mudskippers, the scientists discovers that every year, the growth rate is 0.4% of the population size and the death rate is 5 mudskippers per year.

(i) Write down a differential equation relating M and t. [1]

(ii) Solve the differential equation in part **(i)** and determine what happens to the mudskipper population in the future. [4]

(b) For the crabs, the scientists propose that C and t are related by the differential equation $\frac{\mathrm{d}C}{\mathrm{d}t} = 8C - 0.005C^2$.

(i) Find the number of crabs in the mangrove swamp when the rate of change of C is a maximum. [2]

(ii) Find C in terms of t. Sketch a graph of C against t. [5]

Solution:

(i)

Growth rate $= \frac{0.4}{100} M = 0.004M$

Death rate = 5

$$\frac{\mathrm{d}M}{\mathrm{d}t} = \text{Growth rate} - \text{death rate}$$
$$= 0.004M - 5$$

(ii)

$$\frac{\mathrm{d}M}{\mathrm{d}t} = 0.004M - 5 = 0.004\left(M - 1250\right)$$
$$\frac{1}{M-1250}\frac{\mathrm{d}M}{\mathrm{d}t} = 0.004$$

Integrate wrt t:

$$\int \frac{1}{M-1250}\,\mathrm{d}M = \int 0.004\,\mathrm{d}t$$
$$\Rightarrow \ln|M-1250| = 0.004t + c$$
$$\Rightarrow |M-1250| = \mathrm{e}^{0.004t+c}$$
$$\Rightarrow M - 1250 = \pm\mathrm{e}^{0.004t+c}$$
$$\Rightarrow M - 1250 = A\mathrm{e}^{0.004t}, A = \pm\mathrm{e}^{c}$$

At $t = 0, M = 800$,

$$\Rightarrow A = -450$$
$$\therefore M = 1250 - 450\mathrm{e}^{0.004t}$$

As $t \to \infty$, $450\mathrm{e}^{0.004t} \to \infty$, $M = 1250 - 450\mathrm{e}^{0.004t} \to -\infty$

This means that the mudskipper population will become extinct in the future.

$$M = 1250 - 450\mathrm{e}^{0.004t} = 0 \quad \Rightarrow \quad \mathrm{e}^{0.004t} = \frac{25}{9} \quad \Rightarrow \quad t = 255.4$$

The mudskipper will be extinct in the 256th year.

b(i)

Method 1:

$\dfrac{\mathrm{d}C}{\mathrm{d}t} = 8C - 0.005C^2$.

Let R be the rate of change of C. i.e. $R = \dfrac{\mathrm{d}C}{\mathrm{d}t}$

To maximise R, let $\dfrac{\mathrm{d}R}{\mathrm{d}C} = 0$. Note R is in terms of C.

$$\frac{\mathrm{d}R}{\mathrm{d}c} = \frac{\mathrm{d}\left(\dfrac{\mathrm{d}C}{\mathrm{d}t}\right)}{\mathrm{d}C} = 8 - 0.01C = 0 \quad \Rightarrow \quad C = 800.$$

$$\frac{\mathrm{d}^2R}{\mathrm{d}C^2} = \frac{\mathrm{d}^2\left(\dfrac{\mathrm{d}C}{\mathrm{d}t}\right)}{\mathrm{d}C^2} = -0.01 < 0$$

$\Rightarrow$ The number of crabs when growth rate is maximum is 800.

Method 2:

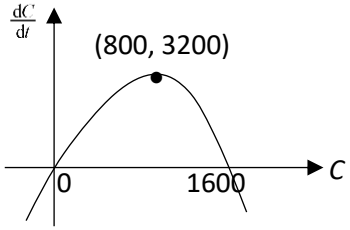
$$\frac{\mathrm{d}C}{\mathrm{d}t} = -0.005\left(C^2 - 1600C\right)$$

$$= -0.005\left(C - 800\right)^2 + 3200 \le 3200$$

$\dfrac{\mathrm{d}C}{\mathrm{d}t} = 3200$ is maximum when C = 800.

Method 3:

$\dfrac{\mathrm{d}C}{\mathrm{d}t} = C\left(8 - 0.005C\right) = 0 \Rightarrow C = 0$ or 1600



From the graph, $\dfrac{\mathrm{d}C}{\mathrm{d}t} = 3200$ is maximum when C = 800.

b(ii)

Method 1: using Partial Fraction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = 8C - 0.005C^2 = 0.005C\left(1600 - C\right)$$

$$\Rightarrow \frac{1}{C\left(1600 - C\right)}\frac{\mathrm{d}C}{\mathrm{d}t} = 0.005$$

Integrate wrt t:

$$\int \frac{1}{C\left(1600 - C\right)}\,\mathrm{d}C = \int 0.005\,\mathrm{d}t$$

$$\Rightarrow \frac{1}{1600}\left[\ln|C| - \ln|1600 - C|\right] = 0.005t + d$$

$$\Rightarrow \ln\left|\frac{C}{1600 - C}\right| = 8t + 1600d$$

$$\Rightarrow \frac{C}{1600-C} = \pm \mathrm{e}^{8t+1600d}$$

$$\Rightarrow \frac{C}{1600-C} = B\mathrm{e}^{8t}, B = \pm \mathrm{e}^{1600d}$$

When $t = 0$, $C = 800$, $B = 1$

$$\therefore C = \frac{1600\mathrm{e}^{8t}}{1+\mathrm{e}^{8t}} \text{ (or } C = \frac{1600}{1+\mathrm{e}^{-8t}}\text{)}$$

Method 2: Complete the square and use MF27

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -0.005\left[(C-800)^2 - 800^2\right]$$

$$\frac{1}{(C-800)^2 - 800^2}\frac{\mathrm{d}C}{\mathrm{d}t} = -0.005$$

$$\int \frac{1}{(C-800)^2 - 800^2}\,\mathrm{d}C = \int -0.005\,\mathrm{d}t$$

$$\frac{1}{2(800)}\ln\left|\frac{(C-800)-800}{C-800+800}\right| = -0.005t + d$$

$$\left|\frac{(C-800)-800}{C-800+800}\right| = \mathrm{e}^{-8t+1600d}$$

$$\frac{C-1600}{C} = \pm \mathrm{e}^{-8t+1600d}$$

$$\frac{C-1600}{C} = B\mathrm{e}^{-8t}, B = \pm \mathrm{e}^{1600d}$$

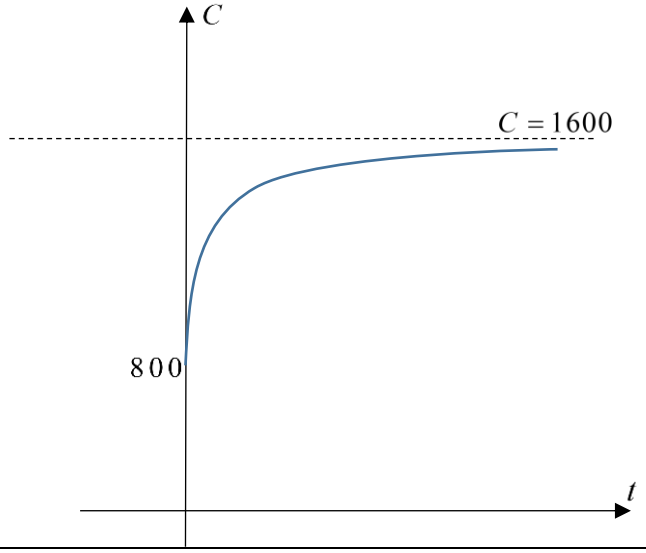
When $t = 0$, $C = 800$

$B = -1$

$$1 - \frac{1600}{C} = -\mathrm{e}^{-8t}$$

$$\frac{1600}{C} = 1+\mathrm{e}^{-8t} \Rightarrow C = \frac{1600}{1+\mathrm{e}^{-8t}}$$

Graph of $C = \dfrac{1600}{1+\mathrm{e}^{-8t}}$



Source of Question: TJC Prelim 9758/2025/01/Q12

10 (a) Show, by means of the substitution $w = x^2 y$, that the differential equation

$$2y + x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^4 y^2}{\sqrt{x^2+1}}$$

can be reduced to the form $\frac{\mathrm{d}w}{\mathrm{d}x} = \frac{w^2 x}{\sqrt{x^2+1}}$. [2]

Hence find the general solution of the differential equation $2y + x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^4 y^2}{\sqrt{x^2+1}}$, leaving your answer in the form $y = \mathrm{f}(x)$. [3]

(b) At a durian plantation, mature durians are susceptible to pests' infection. The spread of pests at the plantation resulted in durians being classified into two categories, either they are bad durians that have been infected by the pests or good durians that have not been infected. It is given that x denotes the number of bad durians, in thousands, in a fixed population size P, in thousands, where $P > 4$. The rate of spread of infection, $\frac{\mathrm{d}x}{\mathrm{d}t}$, where t represents the time taken in days, can be modelled as being proportional to the product of the number of bad durians and good durians that have not been infected. It is given that $x = 4$ and $\frac{\mathrm{d}x}{\mathrm{d}t} = P - 4$ when $t = 0$.

(i) Write down the differential equation involving $\frac{\mathrm{d}x}{\mathrm{d}t}$, x and P [2]

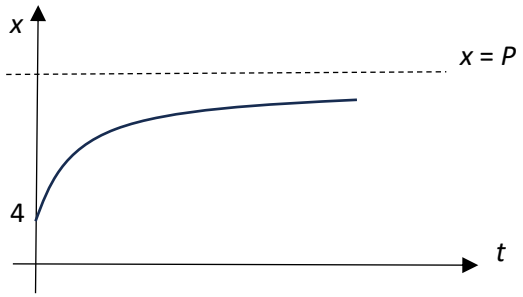
(ii) Solve the differential equation in part **(i)** and find x in terms of P and t. [5]

(iii) By using an appropriate graph, explain in context, the long-term implications if no additional measures are carried out to control the spread of the pests' infection. [2]

Solution:

(a)	Differentiate $w = x^2 y$ with respect to x, $\frac{\mathrm{d}w}{\mathrm{d}x} = 2xy + x^2\frac{\mathrm{d}y}{\mathrm{d}x}$ --------- (1) Multiply x to given D.E., we have $2xy + x^2\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^5 y^2}{\sqrt{x^2+1}}$ ------ (2)

	Subst. (1) into (2) and replace x^4y^2 by w^2 : $\frac{\mathrm{d}w}{\mathrm{d}x}=\frac{w^2x}{\sqrt{x^2+1}}$ $\int\frac{1}{w^2}\,\mathrm{d}w=\int\frac{x}{\sqrt{x^2+1}}\mathrm{d}x$ $\int w^{-2}\,\mathrm{d}w=\frac{1}{2}\int 2x\left(x^2+1\right)^{-\frac{1}{2}}\mathrm{d}x$ $\frac{w^{-1}}{-1}=\frac{1}{2}\frac{\left(x^2+1\right)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)}+C$ where C is arb. constant $-\frac{1}{w}=\sqrt{x^2+1}+C$ $-\frac{1}{x^2y}=\sqrt{x^2+1}+C$ $y=-\frac{1}{x^2\left(\sqrt{x^2+1}+C\right)}$
(b)(i)	$\frac{\mathrm{d}x}{\mathrm{d}t}=kx\left(P-x\right)$ where $k>0$ Given that $x=4$ and $\frac{\mathrm{d}x}{\mathrm{d}t}=P-4$, $P-4=4k\left(P-4\right)$ $k=\frac{1}{4}$ $\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{4}x\left(P-x\right)$
(ii)	$\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{4}x\left(P-x\right)$ $\int\frac{1}{x\left(P-x\right)}\,\mathrm{d}x=\int\frac{1}{4}\,\mathrm{d}x$ $\frac{1}{P}\int\frac{1}{x}+\frac{1}{P-x}\,\mathrm{d}x=\int\frac{1}{4}\,\mathrm{d}t$ $\ln\lvert x\rvert-\ln\lvert P-x\rvert=\frac{P}{4}t+C$ where C is arbitrary constant

	$\ln\frac{x}{P-x}=\frac{P}{4}t+C$ where $P-x>0$ and $x>0$ $\frac{x}{P-x}=A\mathrm{e}^{\frac{P}{4}t}$ where $A=\mathrm{e}^{C}$ $\frac{P-x}{x}=\frac{1}{A}\mathrm{e}^{-\frac{P}{4}t}$ $\frac{P}{x}-1=B\mathrm{e}^{-\frac{P}{4}t}$ where $B=\frac{1}{A}$ $x=\frac{P}{1+B\mathrm{e}^{-\frac{P}{4}t}}$ When $t=0, x=4, \ \frac{P}{4}-1=B$ $x=\frac{P}{1+\left(\frac{P-4}{4}\right)\mathrm{e}^{-\frac{P}{4}t}}=\frac{4P}{4+(P-4)\mathrm{e}^{-\frac{P}{4}t}}$
(iii)	$x=\frac{4P}{4+(P-4)\mathrm{e}^{-\frac{P}{4}t}}$  As $t\to\infty$, $\mathrm{e}^{-\frac{P}{4}t}\to 0$. $\therefore x\to P$ In the long term, all the durians in the plantation will be infected and become bad durians.